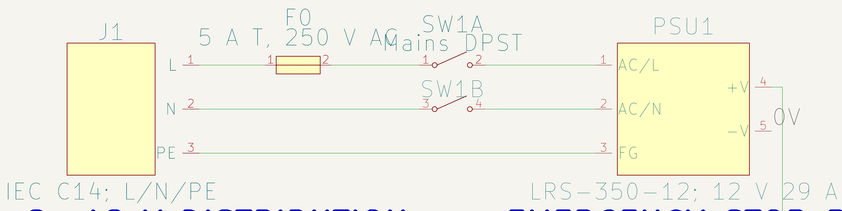
KIRO SO-101 MOBILE DUAL-ARM ROBOT | ELECTRICAL SCHEMATIC (rev F)

1 AC INPUT (mains - unplug before work)

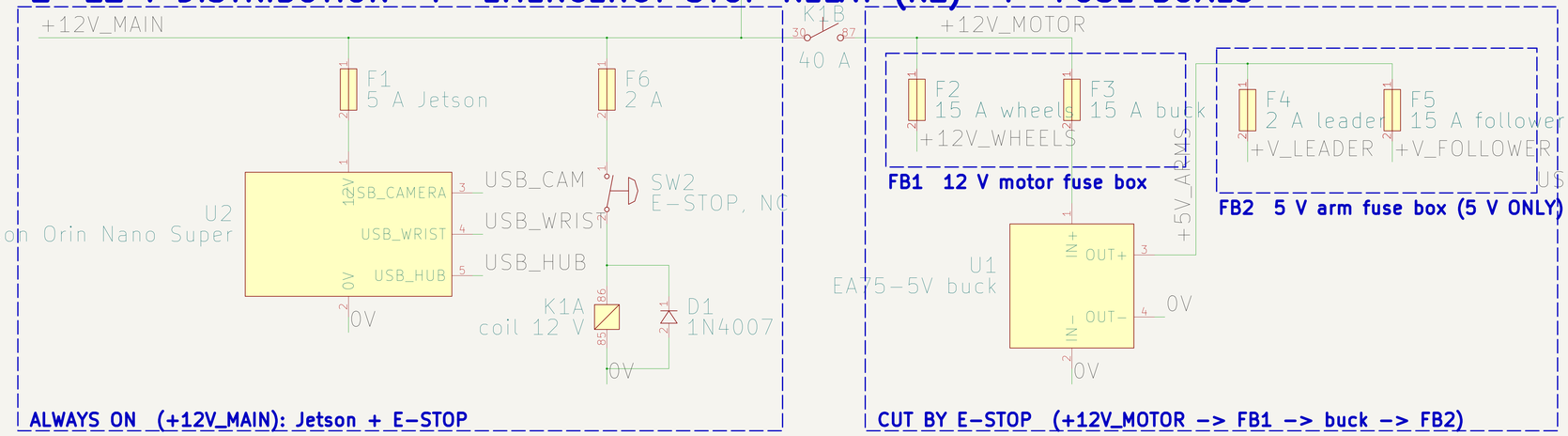


HOW THE E-STOP WORKS

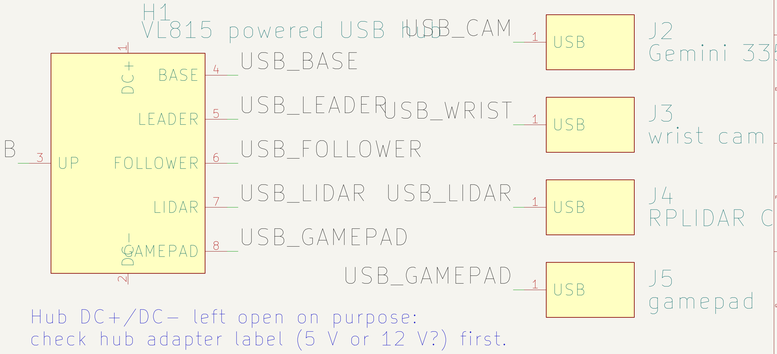
1. Normal run: E-STOP (SW2, NC) is closed, relay K1 coil (86-85) gets 12 V, contact 30-87 closes.
2. Press E-STOP: coil loses power, K1 opens, +12V_MOTOR is dead. FB1 -> wheels and FB1 -> buck -> FB2 -> both arms stop.
3. Jetson (F1) is before K1 on +12V_MAIN and keeps running. Twist E-STOP to release: K1 closes and motor power returns.

K1: 12 V relay, coil 86/85, contact 30 -> 87 rated >= 40 A DC. D1 across the coil (cathode to 86) kills the turn-off spike.
FB1 is fed ONLY from K1 pin 87. F1 (Jetson) and F6 (E-STOP) use their own inline fuse holders on +12V_MAIN.
NEVER connect 12 V to FB2 or an arm adapter: arm servos are limited to 8.0 V.
Software: stop the robot program (torque off) before releasing E-STOP. EA75-5V buck is 15 A TOTAL for both arms.

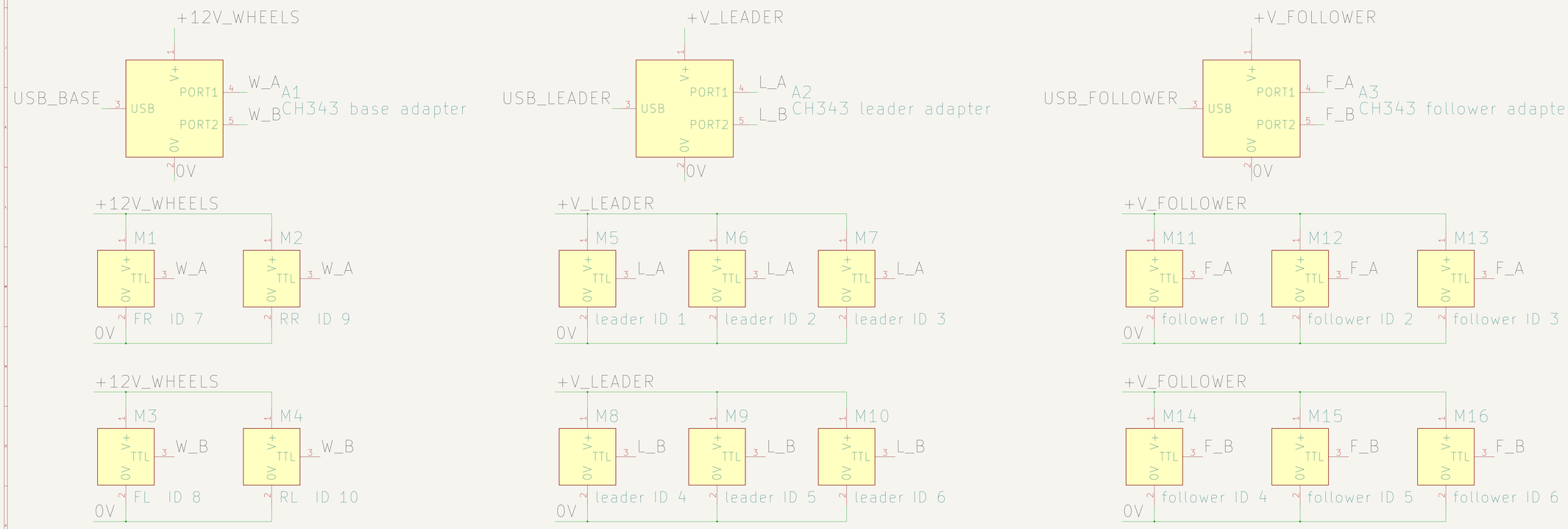
2 12 V DISTRIBUTION + EMERGENCY STOP RELAY (K1) + FUSE BOXES



3 USB LOGIC LINKS



4 WHEELS 12 V (FB1-F2, cut by E-STOP) / LEADER ARM 5 V (FB2-F4, cut by E-STOP) / FOLLOWER ARM 5 V (FB2-F5, cut by E-STOP)



Servo types: wheels ST3215-C018 (12 V); arms STS3215 (5 V rail, 8.0 V limit).

Power, servos and logical USB links: validate against hardware
Rev F: NC E-STOP + relay K1 cut FB1 (wheels, buck) and FB2 (arms); Jetson stays on
Kiro SO-101 project
Sheet: /
File: KIRO_FINAL.kicad_sch
Title: Kiro robot - complete two-arm electrical schematic
Size: A2 | Date: 2026-11-02 | Rev: F
KiCad 8.0.4 10.0.7 | Id: 1/1